

Specific Average Angular Momentum Increase of the Milky Way and Andromeda’s Merged Dark Matter Halo Remnant

LAINÉ KOWALSKI¹

¹*University of Arizona*

Keywords: Galaxy evolution — Dark Matter Halo — Major Merger — Virial radius — Cold Dark Matter Theory — Local Group — Spiral Galaxy

1. INTRODUCTION

A galaxy’s dark matter halo is an invisible spherical cloud of dark matter that surrounds a galaxy, extending far beyond the visible stars, and is crucial for stabilizing the galaxy’s structure. Dark matter halos drive galaxy evolution by dominating the gravitational forces that govern a galaxy’s orbital dynamics. Halos play a critical role in shaping the outcome of galactic mergers, such as for merging dark matter halo components of the Milky Way (MW) and Andromeda (M31) galaxies, which will eventually result in a single, uniform halo.

In terms of galaxy evolution, the process by which galaxies change and develop over time due to star formation, interactions with other galaxies, and accumulation of gas and dark matter, determining the post-merger physical attributes of dark matter halo particles within the combined MW-M31 system is critical for understanding how galaxies merge and behave dynamically. Examining specifically how the dark matter halo components interact during a merge interaction may also help to shed light on the nature of dark matter and its central role within galaxy formation, evolution, and internal galactic interactions. Studying how dark matter halos interact through the merging process helps

astronomers piece together the evolution and future properties of these two galaxies as they become one, which results in a better understanding of galaxy merger dynamics outside of our galactic Local Group as well. Further understanding of dark matter also has broader implications for particle, nuclear, and other areas of contemporary physics.

Regarding dark matter halo dynamics and its role in galaxy evolution, the literature states that dark matter halos are essential galactic structures that surround/host galaxies (J. Dubinski 1992). Galaxies form at the center of dark matter halos, and dark matter halo development/evolution is consistent in galaxies across the universe (C. Frenk & S. White 2012). Therefore, dark matter plays a significant role in shaping all galaxies, regardless of their size or brightness (J. Wolf et al. 2010). Initial central galaxy formation affects the shape and mass of the surrounding dark matter halo because as baryons (normal matter) gather at the center of a halo, the shape of the halo becomes more symmetric and less elongated (M. G. Abadi et al. 2010). Dark matter halos form and grow, starting as small structures that stop expanding, collapse and then merge or gather more dark matter to become larger, in a process known as hierarchical growth (C. Frenk & S. White 2012). Merging halos of similar size, such as

MW and M31, will thoroughly mix their layers (C. Frenk & S. White 2012). In terms of halo mergers, energy involved in the merger and the way halos spin around each other greatly influence the final shape and size of a merged halo. Final merged halos can be elongated or flattened, and the size increases in a predictable way based on the energy and spin from before the merger (N. E. Drakos et al. 2019). Dark matter also has significant implications for a galaxy’s angular momentum. The general trend in the universe is that angular momentum is transferred from central regions to outer edges during galactic mergers (S. Peirani et al. 2004). Therefore, a significant increase in angular momentum at greater distances, especially after a merger, is expected. So, specific average angular momentum should be greater at further radii for merged galaxies than compared to the pre-merger galaxies. Dark matter halos are also categorized into two types: "accretion" halos, which grow smoothly, and "merger" halos, which experience immediate growth by quickly absorbing large amounts of mass. Over time, all halos increase their spin, but halos formed through mergers gain more spin relative to their size than those formed through steady accretion (S. Peirani et al. 2004). When two galaxies merge, they also don’t just combine their internal spin, they bring in orbital angular momentum from their paths around each other. This extra angular momentum gets transferred into the final dark matter halo, meaning the merged halo can spin faster than either of the original halos J. S. Bullock et al. (2001). Finally, the shape and rotation of cosmic halos are influenced by the direction and strength of tidal forces, which primarily cause halos to spin smoothly along specific axes, helping to explain galaxy rotations (J. Dubinski 1992). The Milky Way and Andromeda are on a nearly head-on collision, but they still have a slight impact parameter (they aren’t perfectly aligned). Even

small "misses" can generate torques that increase the spin of the resulting halo H. Hetzner & A. Burkert (2006). Also, a spin parameter (λ), which tells you how much a dark matter halo is spinning compared to how much it could be spinning, is positively correlated with specific angular momentum. So, by calculating the specific angular momentum we can understand how much faster a merged galaxy is spinning or rotating. Higher λ s represent more rotation mostly due to angular momentum M. Vitvitska et al. (2002). Overall, dark matter halos are the static framework for galaxy creation and are the ultimate determining factor for galaxy’s physical and dynamical attributes, predominantly angular momentum and spin.

Several open questions still exist regarding dark matter halo mergers, which are crucial for advancing our understanding of galaxy formation and structure. Key inquiries focus on how these mergers affect the halo mass function over time, the specific properties of dark matter that influence merger dynamics, and the detailed physical processes involved, including the transfer of angular momentum (J. S. Bullock & M. Boylan-Kolchin 2017). Questions remain about the effects of baryonic matter and feedback mechanisms from star formation and black hole activity on these mergers (J. S. Bullock & M. Boylan-Kolchin 2017). In addition, understanding the time it takes for the merged dark matter halos to reach equilibrium is a significant area of ongoing study. These questions highlight the complexity of halo mergers and their fundamental role in cosmological research.

2. PROJECT

This project will investigate the dynamics of the merged dark matter halo versus what the MW halo and M31 halo dynamics were individually before the merger. I will compute the specific average angular momentum of MW, M31 and the merged Milkdrameda dark matter halo

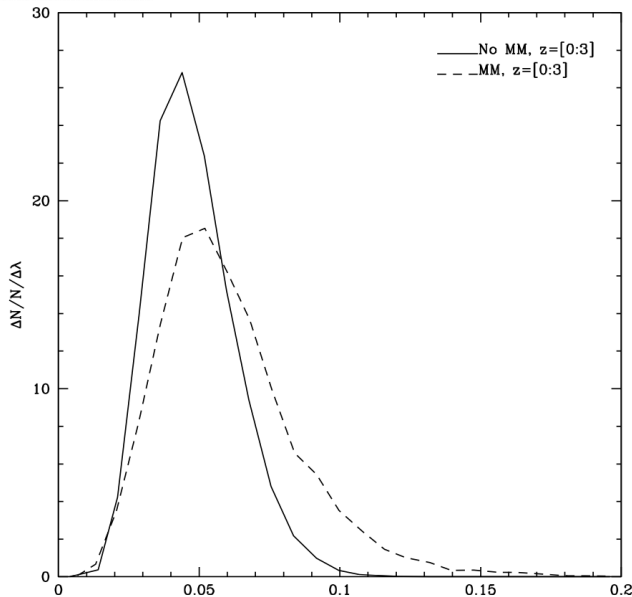


Figure 1. Figure 10 from Ref. (M. Vitvitska et al. 2002) This graph shows how dark matter halos that went through a major merger (dashed line) tend to spin faster than those that didn't (solid line). The dashed line demonstrates higher spin parameter. This rise in spin is a consequence of increased specific angular momentum transferred during mergers, particularly through tangential orbits, as specific angular momentum and the spin parameter are directly correlated. Understanding this angular momentum buildup is essential for explaining why post-merger galaxies often rotate faster.

at different radii graphically to observe where average angular momentum peaks for each. I will observe how the specific average angular momentum differs before the merger, when MW and M31 are individual galaxies, and when the galaxies combine to form Milkdrumeda. These three results will be compared to explore if the profiles are similar/different and the reason why.

Overall, this project addresses questions about the physical processes involved in a dark matter halo merger, and how the MW-M31 merger affects specific average angular momentum of the halo particles, as mentioned in the Introduction. Specifically, I am investigating if the specific average angular momentum magnitude as a function of galactic radius increases

post-merger? And also, is the peak specific average angular momentum at the same or different radii compared to the peak average angular momentum of the dark matter halos in either galaxy before they merged and why?

Answering questions about the physical processes involved in a dark matter halo merger, specifically regarding how the MW-M31 merger affects the specific angular momentum of the halo particles, is critical for several reasons in the context of galaxy evolution. Studying changes in angular momentum due to galaxy mergers helps us to understand the dynamic processes that drive galaxy evolution. Angular momentum is a fundamental property that affects the shape and structure of galaxies, influencing everything from star formation to the distribution of matter. Investigating how the average angular momentum changes across galactic radii post-merger could help infer how mergers reshape galaxies, such as an increase in angular momentum at larger radii might suggest more extended/disrupted galactic disks, impacting the overall structure of the merged galaxy. The location of the peak average angular momentum, depending on if it shifts or remains the same post-merger, provides insights into the effectiveness of theoretical models in predicting the outcomes of major merger events. Also, since dark matter constitutes a significant portion of a galaxy's mass and influences its gravitational field, understanding its role during mergers could reveal more about the nature of dark matter.

3. METHODOLOGY

This project utilizes simulation details from literature by Besla +2012 to investigate changes in specific average angular momentum pre and post merger (G. Besla et al. 2012). The N-body simulation data in this paper models the gravitational interactions of multiple bodies, such as multiple galaxies and different particle types within each galaxy, to understand their dynam-

ics and evolution over time. Before these two galaxies are merged, initially, the Milky Way is a barred spiral galaxy with a central bulge and a supermassive black hole at the center, a prominent stellar disk with spiral arms with a diverse range of stars/matter, and a surrounding, elongated dark matter halo. Andromeda is similar in size to the Milky Way, as it is also a barred spiral galaxy with a central bulge, surrounded by a large stellar disk with spiral arms rich in young stars, gas, and dust. It also has an extensive dark matter halo, and ongoing research is still determining its exact halo shape. Both galaxies' dark matter halos are quite extensive and are most likely slightly elongated. Both also have a virial radius, where the angular momentum peaks, at around 200 kpc. These two galaxies will merge in around 6.7 Gyrs, and are part of our galactic Local Group, along with other satellite and dwarf galaxies.

The project code will analyze the dynamics of the merged dark matter halo compared to the pre-merger dynamics of the Milky Way (MW) and Andromeda (M31) halos. It selects two points in time, before the merger (Snapshot 0) and post-merger (Snapshot 680), calculates the specific average angular momentum magnitude at various radii for MW, M31, and the merged Milky Andromeda, and compares these profiles to understand their differences. From the simulation data, I will only utilize particle type 1, which are dark matter halo particles of all equal mass. Only the LowRes simulation data is needed for this project.

The functions in my code calculate the center of mass position r_{COM} using the equation for center of mass, where R_i is the position vector of particle i , and N is the total number of particles within a maximum radius, R_{max} . To improve accuracy, the code iteratively reduces R_{max} and recalculates r_{COM} until the value converges, or stops changing. This method ensures a stable center of mass for both pre and post merger

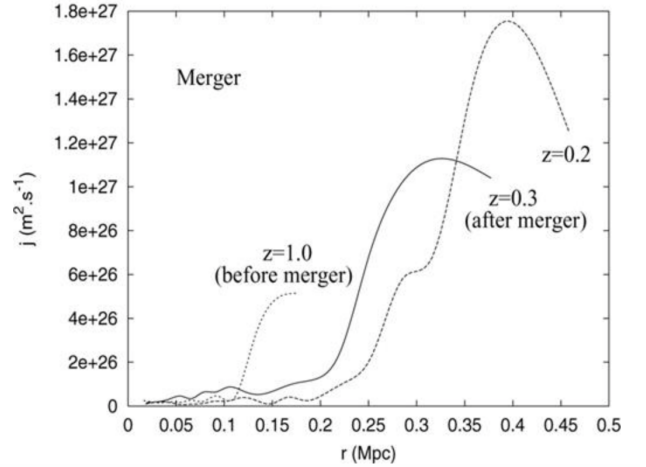


Figure 2. Figure 6 from Ref. (S. Peirani et al. 2004) This graph demonstrates how the average angular momentum of the dark matter halo changes as a function of the radius before and after a merger. The first curve, before merger, represents an arbitrary galaxy's angular momentum as peaking earlier compared to the post merger curve, which peaks at a further radii. This indicates that angular momentum is transferred to a merged galaxy's periphery and generally increases in magnitude once merged, as indicated by the higher peak.

galaxies, which is necessary for calculating angular momentum. To compute the absolute specific average angular momentum L_{ave} at a given radius r , the code calculates the cross product of position and velocity vectors for particles within a radial bin, relative to the system's center of mass. Each term $(\mathbf{r}_i - \mathbf{r}_{COM}) \times (\mathbf{v}_i - \mathbf{v}_{COM})$ gives the specific angular momentum of particle i , which is averaged over all N particles (type 1) in the bin. This equation in the code computes specific average angular momentum at different radii, helping to reveal the rotational structure/dynamical evolution of the dark matter halo.

Three different plots will be generated through the code demonstrating dark matter halo specific average angular momentum at various radii for MW and M31 individually pre-merger and for the merged Milky Andromeda halo.

These plots are scatter plots, where data points represent the magnitude of specific average angular momentum at a specific radius. From these scatter plots, peak specific angular momentum will be calculated at the virial radii of the galaxies to determine if it is a larger magnitude post-merger. Code from Homework6's OrbitCom notebook is also imported to determine where the post-merger snapshot is graphically (past 6.7 Gyrs) and the Homework5 Mass Profiles for Milky Way components graph was utilized to determine the radius at which around 80% of the dark matter halo mass is enclosed for the pre-merger galaxies (around 200 kpc). This graph helped to determine the virial radius utilized code.

The literature states that the transfer of angular momentum, especially after major mergers, moves from the central regions to the periphery and is larger in magnitude (S. Peirani et al. 2004). Therefore, after the dark matter halos of MW and M31 merge into one, I hypothesize that the peak average angular momentum should occur at a further radius (further virial radius) from the center and will have a greater magnitude than compared to the two pre-merger, individual galaxies.

4. RESULTS

Figure 3 shows how the specific average angular momentum of dark matter halo particles in the Milky Way evolves as a function of radius before the galaxy merger occurs. The angular momentum was computed in radial bins extending outward from the Milky Way's center of mass. The resulting scatter plot demonstrates a gradual rise in specific angular momentum with increasing radius, reaching a peak at 3053.82 $\text{kpc} \cdot \text{km/s}$ at the virial radius (200 kpc) before tapering off after. This reflects the rotational structure of the Milky Way's dark matter halo prior to the merger. The Milky Way's specific average angular momentum increases with

radius but remains relatively low, particularly beyond 200 kpc.

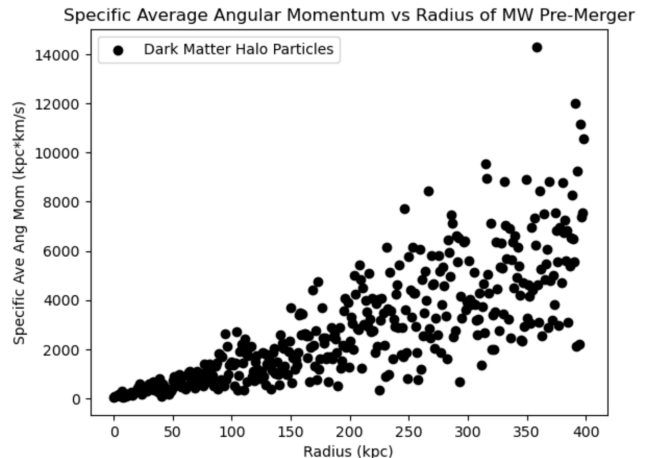


Figure 3. This scatter plot demonstrates the specific average angular momentum of dark matter halo particles in the Milky Way galaxy as a function of radius before the galaxy merger. The plot shows a gradual increase in specific angular momentum with radius, peaking within approximately 200 kpc (virial radius) before flattening out, as the scatter points become more horizontal past this point. The specific angular momentum peak is around 3053.82 $\text{kpc} \cdot \text{km/s}$. The overall magnitude remains relatively low beyond the inner regions, reflecting the structure of the pre-merger Milky Way halo.

The Figure 4 scatter plot presents the radial profile of specific average angular momentum for M31's dark matter halo particles prior to the merger. Like the Milky Way plot, the computation is based on relative positions and velocities with respect to M31's center of mass at snapshot 0. The angular momentum steadily rises across the inner regions, peaking at 2697.34 $\text{kpc} \cdot \text{km/s}$ at the virial radius 200 kpc. The similarity in profile shape between this and the Milky Way's figure supports the idea that both halos exhibit structured but limited/slower rotation or spin before merging. M31's dark matter halo shows a modest rise in specific angular momentum with radius, but values remain relatively low beyond the central regions.

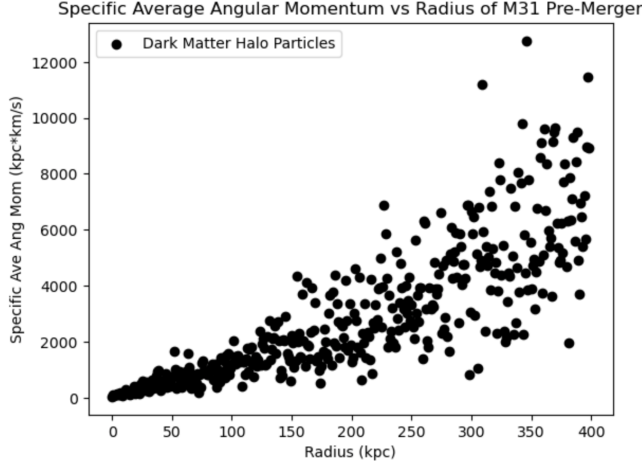


Figure 4. This scatter plot demonstrates the specific average angular momentum of dark matter halo particles in the Andromeda galaxy as a function of radius before the galaxy merger. The plot shows a gradual increase in specific angular momentum with radius, peaking within approximately 200 kpc (virial radius) before flattening out, as the scatter points become more horizontal past this point. The specific angular momentum peak is around $2697.34 \text{ kpc} \cdot \text{km/s}$. The overall magnitude remains relatively low beyond the inner regions, reflecting the general structure of the pre-merger Andromeda halo.

The post-merger scatter plot, Figure 5, illustrates the post-merger rotational structure of the newly formed Milkdromeda dark matter halo, computed at snapshot 680, well after the major merger event. The same analysis was applied by computing average specific angular momentum in radial bins relative to the center of mass of the merged system. Unlike the pre-merger plots, this figure shows a dramatic rise in specific average angular momentum peaking at $14009.52 \text{ kpc} \cdot \text{km/s}$ at a larger virial radius of 350 kpc, with values significantly exceeding those seen in either pre-merger halos. The specific angular momentum of Milkdromeda is calculated to be around 4.5 times that of the MW, or a 21 percent increase. The specific angular momentum of Milkdromeda is also calculated to be around 5 times that of the MW, or a 19 percent increase. This reflects the redistribu-

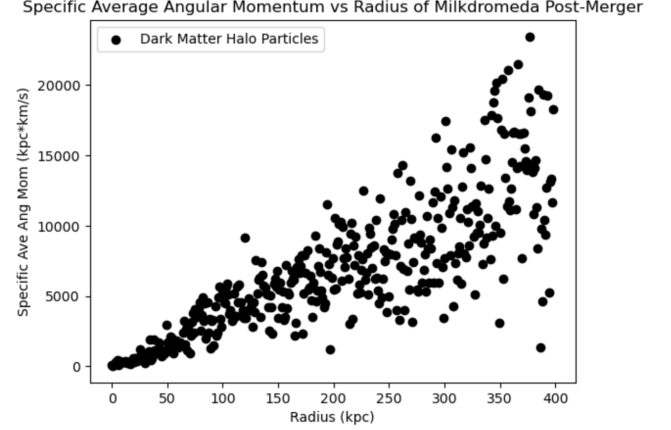


Figure 5. This scatter plot demonstrates the specific average angular momentum of dark matter halo particles in the Milkdromeda galaxy as a function of radius after the galaxy merger. The plot shows a gradual increase in specific angular momentum with radius, peaking within approximately 350 kpc (estimated virial radius) before flattening out, as the scatter points become more horizontal past this point. The specific angular momentum peak is around $14009.52 \text{ kpc} \cdot \text{km/s}$. This specific angular momentum is around 4.5 times greater than compared to the momentum values for the individual pre-merger galaxies. Therefore, the plot demonstrates that merged halos have a greater specific angular momentum magnitude and at greater radii.

tion of angular momentum outward during the merger process, most likely driven by torques and dynamical friction. Therefore, after the major merger, Milkdromeda's dark matter halo exhibits much higher specific average angular momentum at greater radii, confirming that mergers can significantly enhance outer halo rotation.

5. DISCUSSION

The comparison between Figures 3, 4, and 5 clearly demonstrates how the specific average angular momentum of the Milky Way and M31 dark matter halos evolved before and after the major merger. Figures 3 and 4 show that, prior to the merger, both galaxies' dark matter halos showed only a modest increase in specific aver-

age angular momentum with radius, with peaks occurring at 200 kpc (virial radii). At this radius, dark matter halo particle density is sufficient enough to call this the peak, at around $3000 \text{ kpc} \cdot \text{km/s}$ for MW and M31 individually. In contrast, Figure 5, which represents the post-merger Milkdromeda dark matter halo, reveals a dramatic increase in specific average angular momentum at much larger radii, with values far exceeding those observed in either the Milky Way or M31 individually. At around the virial radius, 350 kpc, the specific angular momentum is around $14000 \text{ kpc} \cdot \text{km/s}$, demonstrating the specific angular momentum is much greater than MW and M31. The specific angular momentum of Milkdromeda is calculated to be around 4.5 times that of the MW, or a 21 percent increase. The specific angular momentum of Milkdromeda is also calculated to be around 5 times that of the MW, or a 19 percent increase. This increased magnitude at greater distances strongly supports the hypothesis that major mergers redistribute angular momentum outward, enhancing the rotation of the outer halo regions. These results align with my hypothesis that angular momentum increases in magnitude with radius towards the periphery of the merged galaxy, compared to the individual galaxies pre-merger. Therefore, the merged galaxy has greater overall spin/rotation as predicted.

The results align with theoretical expectations that major mergers transfer angular momentum from the central regions to the outskirts, causing the outer, merged halo to spin faster and exhibit a higher magnitude of specific angular momentum (S. Peirani et al. 2004). This confirms that major mergers produce greater rotation/spin due to increased specific angular momentum M. Vitvitska et al. (2002). Torques and dynamical interactions during the merger explain why the merged galaxy spins more rapidly and achieves greater angular momentum values

at larger distances from the center of mass (J. Dubinski 1992). The literature further states that the rotation of halos is influenced by the direction and strength of tidal forces (J. Dubinski 1992). Since the Milky Way and Andromeda are not perfectly aligned, small offsets during their merging process generated torques that contributed to this increased spin observed in my results H. Hetznecker & A. Burkert (2006). Merged halos spin faster than either of the original halos due to the additional orbital angular momentum introduced during the merger, consistent with what my results demonstrated J. S. Bullock et al. (2001). My findings also support the conclusion that dark matter halos formed through mergers experience an increase in spin relative to their size (S. Peirani et al. 2004). The specific angular momentum more than quadrupled, and doubled in size for the merged galaxy compared to the pre-merger systems, reinforcing the idea that the dark matter halo became thoroughly mixed (C. Frenk & S. White 2012). Additionally, the combination of dark matter particles to form the merged halo illustrates the broader concept of how galaxies evolve through hierarchical growth (C. Frenk & S. White 2012). This result shows that major mergers play a crucial role in redistributing and boosting the angular momentum of dark matter halos, especially at larger radii. It highlights how mergers not only grow galaxies in size but also fundamentally reshape their rotational structure, driving the hierarchical evolution of galaxies over cosmic time.

However, there are several uncertainties in this analysis. Small errors can arise when calculating the center of mass and relative velocities, and the choice of radial bin size and simulation resolution can influence the measured angular momentum profiles. The assumption of specific angular momentum more than quadrupling after the merger is also a large simplification if there are slight code calculation errors, and ex-

cluding baryonic matter means the results do not fully represent the real galaxy dynamics of the merged halo. I also roughly estimated the virial radii of MW and M31 to be around 200 kpc and Milkdrameda's to be 350 kpc. Despite

these uncertainties, the results clearly support the idea that major mergers significantly enhance the angular momentum of dark matter halos and play a key role in shaping the structure and evolution of galaxies.

REFERENCES

- Abadi, M. G., Navarro, J. F., Fardal, M., Babul, A., & Steinmetz, M. 2010, *Monthly Notices of the Royal Astronomical Society*, 407, 435, doi: [10.1111/j.1365-2966.2010.16912.x](https://doi.org/10.1111/j.1365-2966.2010.16912.x)
- Besla, G., Kallivayalil, N., Hernquist, L., et al. 2012, *Monthly Notices of the Royal Astronomical Society*, 421, 2109, doi: [10.1111/j.1365-2966.2012.20466.x](https://doi.org/10.1111/j.1365-2966.2012.20466.x)
- Bullock, J. S., & Boylan-Kolchin, M. 2017, *Annual Review of Astronomy and Astrophysics*, 55, 343, doi: [10.1146/annurev-astro-091916-055313](https://doi.org/10.1146/annurev-astro-091916-055313)
- Bullock, J. S., Dekel, A., Kolatt, T. S., et al. 2001, *ApJ*, 555, 240, doi: [10.1086/321477](https://doi.org/10.1086/321477)
- Drakos, N. E., Taylor, J. E., Berrouet, A., Robotham, A. S. G., & Power, C. 2019, *Monthly Notices of the Royal Astronomical Society*, 487, 993, doi: [10.1093/mnras/stz1306](https://doi.org/10.1093/mnras/stz1306)
- Dubinski, J. 1992, *ApJ*, 401, 441, doi: [10.1086/172076](https://doi.org/10.1086/172076)
- Frenk, C., & White, S. 2012, *Annalen der Physik*, 524, 507, doi: <https://doi.org/10.1002/andp.201200212>
- Hetznecker, H., & Burkert, A. 2006, *Monthly Notices of the Royal Astronomical Society*, 370, 1905, doi: [10.1111/j.1365-2966.2006.10631.x](https://doi.org/10.1111/j.1365-2966.2006.10631.x)
- Peirani, S., Mohayaee, R., & de Freitas Pacheco, J. A. 2004, *Monthly Notices of the Royal Astronomical Society*, 348, 921, doi: [10.1111/j.1365-2966.2004.07412.x](https://doi.org/10.1111/j.1365-2966.2004.07412.x)
- Vitvitska, M., Klypin, A. A., Kravtsov, A. V., et al. 2002, *Astrophysical Journal*, 581, 799, doi: [10.1086/344361](https://doi.org/10.1086/344361)
- Wolf, J., Martinez, G. D., Bullock, J. S., et al. 2010, *MNRAS*, 406, 1220, doi: [10.1111/j.1365-2966.2010.16753.x](https://doi.org/10.1111/j.1365-2966.2010.16753.x)